

MATHEMATICS FINAL EXAM 2024

Duration: 2 hours | Total Marks: 50

1. What is the derivative of x^2 ?

- A. x
- B. $2x$
- C. x^2
- D. $2x^2$

2. The integral of $1/x$ is:

- A. $\ln(x) + C$
- B. $x + C$
- C. $1/x^2 + C$
- D. $x^2 + C$

3. What is the value of $\sin(90^\circ)$?

- A. 0
- B. 0.5
- C. 1
- D. -1

4. If $f(x) = 3x + 2$, what is $f(5)$?

- A. 15
- B. 17
- C. 13
- D. 20

5. The Pythagorean theorem states:

- A. $a + b = c$
- B. $a^2 + b^2 = c^2$
- C. $a * b = c$
- D. $a^2 - b^2 = c^2$

6. What is the area of a circle with radius 5?

- A. 25?
- B. 10?
- C. 5?
- D. 50?

7. The quadratic formula is used to solve equations of degree:

- A. 1
- B. 2
- C. 3
- D. 4

8. What is $\log_{10}(100)$?

- A. 1
- B. 2
- C. 10
- D. 100

9. In a right triangle, the longest side is called:

- A. Hypotenuse
- B. Adjacent
- C. Opposite
- D. Base

10. What is the slope of the line $y = 3x + 7$?

- A. 7
- B. 3
- C. 10
- D. -3

ANSWER KEY

1-B 2-A 3-C 4-B 5-B

6-A 7-B 8-B 9-A 10-B